A method for the grading and recognition of osteoarthritis X-ray images based on the MED-HED detection

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Abstract—**The manual Kellgren-Lawrence (KL) grading of osteoarthritis (OA) X-ray images is prone to inter-rater variability owing to the subjective nature of the evaluation process. In this study, we constructed an automatic KL grading system to realize intelligent assessment of OA severity in X-ray images through preprocessing, segmentation, feature extraction, and classification. The primary steps are as follows: First, preprocessing: This step integrates wavelet denoising, median filtering, and Medical_HED (MED-HED) edge detection to enhance image contrast and extract contours of key knee joint regions. Second, regional segmentation: The region of interest (ROI) is intercepted based on joint space localization, and the images of key regions of the knee joint are obtained by combining the connected region labeling method. Third, feature extraction: Fourier descriptors are employed to convert joint contours into shape vectors, facilitating extraction of geometric features that reflect OA severity. Fourth, grading classification: A template matching framework is constructed to compare shape vector similarity between samples and KL-grade templates, enabling automatic grading. Experimental results show that the system achieves an overall grading accuracy of 92.3% in 400 knee X-ray tests, outperforming conventional operators such as Canny and LOG. The proposed recognition method mitigates subjectivity in manual evaluation through multi-step image processing and shape analysis. Its high accuracy and efficiency make it a reliable computer-aided decision-making tool for clinical OA diagnosis, with potential to replace manual grading.**

Keywords—KL, MED-HED, X-ray images, osteoarthritis

I. INTRODUCTION

Osteoarthritis (OA) is the most prevalent degenerative joint disorder globally, with its incidence rising with age. It has become a major cause of functional impairment and disability among middle-aged and elderly populations. As the most common site of OA, assessing knee joint severity is crucial for clinical diagnosis. The Kellgren-Lawrence (KL) grading system [1], proposed in 1957, has become the gold standard for evaluating OA in clinical X-ray images owing to its ease of operation and low cost. However, outcome evaluation is inherently subjective, as it is influenced by individual perceptions of attending physicians, leading to substantial variability in assessment consistency among different clinicians. This subjectivity may compromise the validity of clinical studies and hinder large-scale screening and precise treatment implementation.

In recent years, artificial intelligence has emerged as a novel solution to this issue. Convolutional neural networks (CNNs) mark a significant advancement in image classification, leveraging automated feature extraction to improve classification accuracy and efficiency [2]. A CNN model pre-trained on ImageNet achieved 74.81% multi-classification accuracy [3], representing a substantial improvement over conventional manual feature methods. The Holistically-Nested Edge Detection (HED) algorithm, a classical deep learning-based edge detection model proposed in 2015, shows promising prospects in medical imaging applications [4, 5]. At the feature extraction level, Fourier descriptors (FDs) convert joint boundaries into shape vectors invariant to rotation, translation, and scaling [6, 7]. In the classification stage, the dynamic time warping (DTW) algorithm establishes a template matching framework to quantify shape vector similarity between test samples and KL-grade templates, enabling automatic classification [8, 9]. Despite the research conducted on osteoarthritis X-ray images as outlined above, it remains challenging to accurately capture key pathological features, such as joint space narrowing and bone redundancy morphology, in complex noise environments.

To address the issues of strong subjectivity and low efficiency in manual assessment, this study develops a grading and recognition method for osteoarthritis X-ray images based on Medical_HED (MED-HED). Through steps such as preprocessing, regional segmentation, feature extraction, and classification, intelligent evaluation of OA severity in knee joint X-ray images is achieved, enabling automatic grading. The results demonstrate the effectiveness of the proposed method. This research can improve the objectivity and efficiency of diagnosis, provide a reliable computer-aided decision-making tool for clinical OA diagnosis, and thus holds significant clinical application value.

II. METHOD

A. Algorithmic framework

The workflow for processing X-ray images is illustrated in Figure 1. Notably, the algorithm for extracting bone joint gap regions consists of two components: first, using the connected region labeling method; second, applying an enhanced Fourier descriptor algorithm to extract shape vectors of bone joint contour lines from X-ray image samples. A classification algorithm then aligns these shape vectors with those in a bone joint X-ray image template library to assess joint conditions.

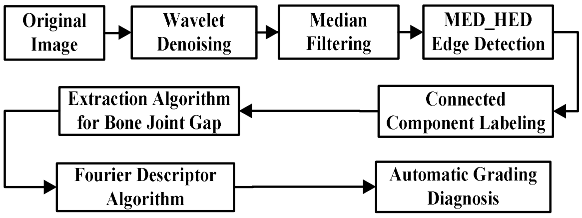


Figure 1 X-ray image processing flow

The X-ray image grading recognition algorithm involves the following steps:

1）Filtering bone and joint radiographs using wavelet denoising and median filtering.

2）Extracting X-ray image contours via the MED-HED edge detection operator.

3）Intercepting contours of key regions using a neighborhood feature algorithm and optimizing them with connected region labeling.

4）Extracting feature vectors from key region contours using a modified Fourier descriptor algorithm.

5）Applying a feature matching classification algorithm to align feature vectors with template libraries for KL grade identification.

Wavelets are widely used in denoising processes. The wavelet filter coefficient is modulated by adjusting the threshold value. Fluctuations that exceed this coefficient are recognized as the original image signal, while those that fall below it are considered irrelevant interference. Consequently, the selection of the threshold value is of paramount importance. Thresholds can be categorized into two distinct types: global and local. The selection of an appropriate threshold is crucial because improper selection can lead to undesirable outcomes. In instances where the threshold is set too high, significant distortions may occur. Conversely, if the threshold is set too low, the filtering effect may be insufficient.

Median filtering is a nonlinear filtering technique that enhances the edge-smoothing denoising process. It employs the theory of sorting statistics to eliminate isolated noise points, thereby improving the overall quality of the image. The principle of median filtering involves sorting all pixels within a specified field according to their size, followed by the selection of the appropriate gray value.

B. MED-HED

The edge detection operator employs the enhanced MED_HED method. HED, a classical deep learning edge detection model proposed in 2015, is based on the core design idea of realizing high-precision edge detection through multi-scale feature fusion and deep supervision mechanism. This study aimed to propose a MED-HED that targets the pathological features of osteoarthritis. These features include irregular edges of bone, cumbersome and fuzzy boundaries of cartilage, wear and tear, and bone cortical fracture under noise interference. The MED-HED is proposed to achieve this objective by making targeted improvements to the original HED network. Through structural optimization, loss function improvement, and feature enhancement, the accuracy of edge detection in complex medical scenarios is enhanced. The specific enhancements are listed below:

(1) Network structural improvements

The backbone network employed ResNet50 instead of VGG16. Residual connections have been demonstrated to effectively mitigate the phenomenon of deep network gradient vanishing. This makes it a suitable approach for the low-contrast edge learning of bones and soft tissues in X-rays. The fully connected layer was removed, and the first four residual blocks (Conv1-Conv4) were retained to output a multi-scale feature map. Furthermore, the channel-weighted feature maps undergo two optimizations via the spatial attention mechanism to ensure precise identification of the joint region in the X-ray image. This module has been demonstrated to effectively suppress the interference of irrelevant background regions, such as muscle and fat, while concomitantly achieving a substantial enhancement in the clarity of feature expression in key regions.

(2) Loss function improvements

In the enhanced MED-HED network, the optimization class balances the Dice loss function, the structural similarity loss function, and the total loss function. The three functions have been developed to address the fundamental challenges associated with X-ray image edge detection, namely, sample imbalance, edge structure blurring, and multi-scale feature fusion.

- Dice loss function

In medical images, the percentage of edge pixels is extremely low, and the direct use of traditional cross-entropy leads to a model that tends to predict the background and is prone to "missed detection". Class Balancing Dice loss is optimized by:

$$L_{Dice} = 1 - \frac{2\sum(y \cdot \hat{y})}{\sum y^2 + \sum \hat{y}^2} + \beta L_{BCE} \tag{1}$$

where y is the edge label, $\hat{y}$ is the prediction probability, and β is the edge pixel gradient weight. This loss function has been demonstrated to optimize both the recall and classification confidence of the edge pixels, thereby reducing leakage and false detection.

- Structural Similarity Index (SSIM)

In X-ray images, the edges of bone and cartilage typically exhibit continuous geometric properties, such as the straight edges of bone cortex and the fractal edges of bone redundancy. Conventional loss functions primarily focus on pixel-level classification, which can result in the formation of broken edges. SSIM loss compels the model to learn the spatial continuity of the edges by calculating the luminance, contrast, and structural similarity between the predicted and actual edges in X-ray images. This approach enables the model to capture the structural characteristics of continuous edges. SSIM loss is introduced to maintain the spatial continuity of edges:

$$L_{SSIM} = 1 - SSIM(y, \hat{y}) \tag{2}$$

SSIM Loss is predicated on SSIM metrics that have been meticulously engineered to ensure spatial continuity and structural coherence of edges.

●Total loss function

$$L_{Total} = L_{fuse} + 0.5\sum_{k=1}^{4} L_k^{Dice} + 0.3L_{SSIM} \tag{3}$$

where L_{fuse} is the loss of the global output, that is, the class-balanced Dice loss of the final fused edge probability map with labels, representing the final edge prediction quality of the model after fusion of multi-scale features. $\sum_{k=1}^{4} L_k^{Dice}$ is the sum of the class-balanced Dice loss (weight 0.5) of the 4 side outputs, which forces the model to learn edge features at different scales through "deep supervision", for example, shallowly capturing the subtle boundaries of cartilage and deeply capturing the overall contour of the bony cambium. L_{SSIM} is structural similarity loss (weight 0.3) to ensure structural continuity of the final edges, for example, straight line features of the bone cortical edges.

C. Connected area marking method

The connected region labeling method defines the connected region from the pixel perspective. In this study, we adopted eight-neighborhood as the core of connectivity region marking method, and the recognition process is as follows:

(1) Determine the leftmost, upper-left, uppermost, and upper-right points in the eight-neighborhood of this point. If none of the points are present, it indicates the start of a new region.

(2) If this point is the leftmost point in the eight-neighborhood of this point, and the upper right are both points, mark this point as the smallest marked point of these two, and modify the large mark to the small mark.

(3) If there is a point to the upper left and a point to the upper right in the eight neighborhoods of this point, mark this point as the smallest marked point of the two, and modify the large mark to a small mark.

(4) Otherwise mark this point as one of four in the order of leftmost, upper left, uppermost, upper right.

D. Partial extraction algorithm for bone joint gap

In order to ascertain the severity level of osteoarthritis disease, it is imperative to extract the osteoarthritic gaps with core regions to further recognize the shape and determine the condition. The number of white pixels in each row was determined and subsequently evaluated against a predetermined threshold. In the event that the number of white pixels in a given row exceeds the specified threshold, and the white pixels in said row exceed those in the subsequent row, yet are fewer than those in the preceding row, the presence of a gap is then identified. It is important to note that the existence of certain interferences can be utilized to employ the connected region labeling method first, followed by the extraction of the joint gap feature part. The following is the extraction algorithm for extracting the joint gap feature:

Step1: Calculate the number of white pixels per row in the image matrix $h(i)$ （ $i = 1,2,...,m$. m is the matrix row paradigm);

Step2: If $h(i) \geq H$ (H is threshold) and $h(i-1) < h(i) < h(i+1)$, mark i ;

Step3: Define the intercept height k_1 and replace the white pixels with black pixels in the rows other than $[i - \frac{k_1}{2}, i + \frac{k_1}{2}]$;

Step4: Truncate all rows and columns that do not contain white pixels to extract the joint gap portion.

E. Principles of the Fourier Descriptor Algorithm

The discrete Fourier transform of the boundary can be used as a basis for quantitatively describing its shape. The principle of using the Fourier descriptor is to transform a two-dimensional image into a one-dimensional vector. That is, a curve segment in the xy-plane is transformed into a one-dimensional function f(r) (in the r-f(r) plane), and also xy can be made to coincide the plane with the complex plane uv, where the real part of the u-axis coincides with the x-axis, and the imaginary part of the v-axis coincides with the y-axis. This can be used to represent each point (x,y) on a given boundary in the form of a complex u+jv. These two representations are essentially the same and are point correspondences. For a sequence of K points in the xy plane， if $x(k) = x_k$、$y(k) = y_k$, a sequence of coordinates is

$$s(k) = x(k) + jy(k) \quad (k = 0,1,...,K-1) \tag{4}$$

The discrete Fourier transform of $s(k)$ can be written as:

$$s(u) = \sum_{k=0}^{K-1} S(k) e^{-j2\pi uk/K} \tag{5}$$

where the Fourier coefficients $s(u)$ can be referred to as the Fourier descriptors of the boundary, which is Fourier inverted as

$$s(k) = \frac{1}{k}\sum_{u=0}^{K-1} s(u) e^{j2\pi uk/K} \tag{6}$$

Thus only the first M coefficients of $s(u)$ are utilized to reconstruct the original image, so that an approximation to $s(k)$ can be obtained without changing its basic shape, i.e.

$$\hat{s}(k) = \frac{1}{M}\sum_{u=0}^{M-1} s(u) e^{j2\pi uk/K} \tag{7}$$

Therefore, after matching the sample Fourier transform with the template Fourier transform, the sample corresponding to the template type can be determined by selecting the one with the smallest gap. Using the improved Fourier descriptor algorithm, the feature vector extraction of the contour line of the key part includes the following steps:

1) Set the size of the descriptor;

2) Set the offset center of the osteoarthritis X-ray image contour line drawing;

3) The Fourier expansion of the contour lines of the osteoarthritis X-ray images of the extracted key areas and the two-dimensional data are transformed into one-dimensional feature vector data.

III. RESULTS

A clinical dataset containing 400 X-ray images of the knee with osteoarthritis was utilized in this study, of which 300 cases were allocated for model training and 100 cases for independent testing. The images were obtained from the radiology departments of three hospitals. The images covered KL grades 0 (normal) to 4 (severe osteoarthritis OA). The case inclusion criteria followed the American College of Rheumatology classification criteria for osteoarthritis. The image resolution was 1024×1024 pixels, encompassing the distal femur, proximal tibia, and patellar regions of the knee. The clinical annotation was conducted independently by three radiologists with over a decade of experience. The consensus value after consistency testing (kappa=0.82) was employed as the gold standard for the final grading results.

The EC edge integrity value indicates the degree to which the detected edges correspond to the actual edges, that is, the proportion of real edges that are accurately identified. A higher EC value corresponds to a lower leakage rate of the algorithm to real edges. The EC calculation formula is as follows.

$$EC = \frac{TP}{FN + TP} \times 100\%$$

where TP (True Positive) is the number of detected true edge pixels and FN (False Negative) is the number of undetected true edge pixels.

The LA value is a measure of how close the detected edge position is to the real edge, i.e., the proportion of detected edges that are real edges. The higher the LA value, the lower the false detection rate of the algorithm. The formula is as follows:

$$LA = \frac{TP}{FP + TP} \times 100\%$$

where FP (False Positive) is the number of falsely detected non-edge pixels.

The original X-ray images were subjected to a series of image processing techniques, including wavelet denoising (decomposition layer 3, soft thresholding), median filtering (3 × 3 window), and edge extraction using the Canny operator, LOG operator, and improved MED-HED algorithm.

The minimum vertical distance between the medial and lateral femoral condyles and tibial plateau was calculated by the joint gap localization algorithm. The region of interest was intercepted, and the connected region labeling method was used to remove extraneous noise. The contour image of the femoral condyles-tibial plateau contact surfaces of the knee joint was retained. The KL grade template library contains the average shape vectors of 60 training samples for each grades 0-4. As illustrated in Table 1, the EC values and LA values of MED-HED demonstrate superiority over those of the Canny and LOG conventional operators, thereby substantiating its efficacy in the detection of degenerative joint edges in complex scenarios.

As illustrated in Table 2, the edge detection results of the three operators in images with different KL grades are demonstrated. The Canny and LOG operators demonstrate reasonable performance in KL=0 (normal joints). However, as the KL grades increase (e.g., osteochondral hyperplasia and articular facet deformity in KL=3-4), there is a prevalence of missed detection of the osteochondral boundaries and interference of the soft-tissue artifacts, among other issues. The MED-HED algorithm, through multi-scale feature fusion, demonstrates significant superiority over conventional operators in the context of subchondral osteosclerotic bands and irregular bony cumbersomening. It exhibits notable efficacy in enhancing extraction completeness and accurately captures the subtle contour differences between articular surface collapse and bony cumbersomening, particularly in cases of severe degeneration (KL=4 grade).

Table 3 focuses on the edge details of the key parts of the knee joint, showing the localization accuracy advantage of MED-HED in the femur-tibia contact surface and the tip of the bone remnant. In suspected cases, MED-HED can clearly delineate the elevated periphery of the diminutive bone remnant, while the Canny operator engenders indistinct edges owing to its sensitivity to noise. In the well-defined bone remnant, the LOG operator loses the serrated edges of the bone remnant owing to the excessive smoothing of the Gaussian kernel, while MED-HED retains the complete geometrical details, providing precise boundary information for subsequent shape analysis.

As illustrated in Figure 2, the impact of the three edge detection operators on the performance of the grading system was evident. The MED-HED operator demonstrated a higher level of overall accuracy in the test set when compared to the Canny and LOG operators. The MED-HED system exhibited the greatest discriminative power, which was attributed to its reliable extraction of key pathological features, such as subchondral osteosclerosis and joint space narrowing.

Table 1 Comparison of different edge detection operators

	EC	LA
Canny	0.68±0.06	0.72±0.05
LOG	0.65±0.07	0.69±0.06
MED-HED	0.91±0.03	0.94±0.04

Table 2 Bone joint edge detection results

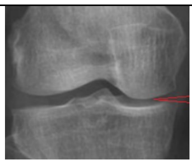
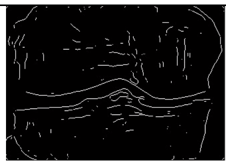
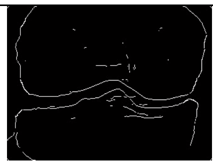
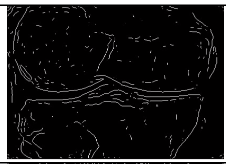
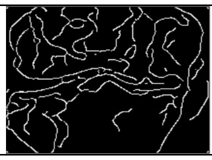
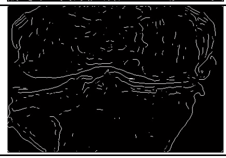
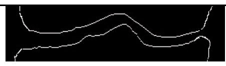
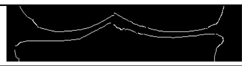
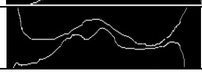
	Original images	Canny	LOG	MED_HED
KL=0				
KL=1				
KL=2				
KL=3				
KL=4				

Table 3 Bone joint Interstitial part extraction results

	KL=0	KL=1	KL=2	KL=3	KL=4
Canny					
Log					
MED-HED					

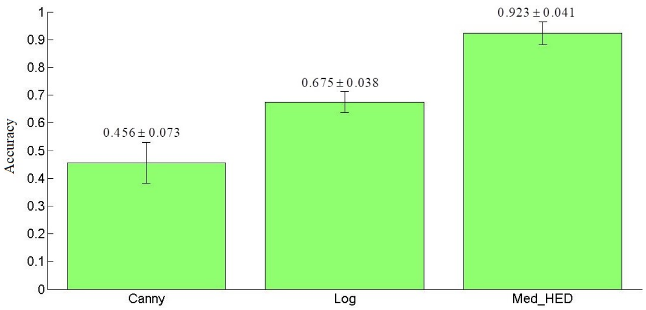


Figure 2 Comparison of recognition grading accuracy of three edge detection operators

IV. Discussion

This study proposes an automatic grading system for osteoarthritis X-ray images, termed MED-HED, which effectively addresses the subjectivity inherent in traditional KL grading. This system integrates image processing and shape analysis techniques to achieve a more objective evaluation. The system demonstrated a grading accuracy of 92.3%.

The complexity of X-ray image recognition stems from the necessity of edge detection, a task in which the MED-HED algorithm has been shown to exhibit notable advantages. In the context of edge detection for complex degenerate joints, the MED-HED algorithm has demonstrated edge integrity and localization accuracy that significantly surpass those of the conventional Canny and LOG operators. This enhancement in performance can be attributed to two optimizations:

●Network structure optimization: ResNet50 was utilized as the primary network rather than VGG16. The issue of gradient disappearance in deep networks is addressed by residual connectivity, and the spatial attention mechanism is incorporated to concentrate on the joint region, thereby effectively suppressing the interference of background noise such as muscle and fat. The design under consideration has been demonstrated to facilitate a substantial enhancement in the precision of extraction of irregular edges of bony cords and subchondral osteosclerotic bands in cases classified as KL=3-4 grade.

●Loss function optimization: The joint optimization of the class-balanced Dice loss and SSIM loss has been demonstrated to reduce leakage while maintaining the spatial continuity of the edges. This reduction in leakage significantly decreases the risk of early lesion leakage.

At the clinical level, this high-precision edge detection provides a reliable basis for subsequent joint gap localization and shape analyes. Research has demonstrated a direct correlation between the completeness of joint edge extraction and the grading system's capacity to differentiate between early osteochondromas and severe joint deformity.

In this study, the enhancement of the grading performance was realized through the joint action of each link. The integration of wavelet denoising and median filtering is effective in the suppressing noise in X-ray image images, thereby generating a pristine background conducive to subsequent edge detection. The region of interest interception, founded upon joint gap localization and the connected region labeling method, ensures the accurate extraction of the key regions of the knee joint while effectively avoiding the interference of irrelevant regions, such as the patella and soft tissues. The Fourier descriptor transforms the joint boundaries into shape vectors that are invariant to rotation, translation, and scaling. Furthermore, its robustness to noise and contrast variations is significantly enhanced.

Nevertheless, this study has some limitations. The present feature extraction is exclusively based on two-dimensional contours and does not consider three-dimensional parameters, including joint space width and bone volume. CT three-dimensional reconstruction technology can quantify joint space width and bone volume. In the future, the potential exists to explore the fusion analysis of CT images. In the future, multimodal features may be integrated to construct a feature space encompassing grayscale, texture, geometry, and other multidimensional characteristics.

V. CONCLUSION

The automatic KL grading diagnosis system for OA encompasses a broad array of disciplines, including orthopedic medicine, image processing, applied mathematics, artificial intelligence, recognition algorithm, and other related fields. This system represents a pioneering advancement in the domain of orthopedic automatic discrimination. The preceding orthopaedic automatic diagnosis field has conducted pertinent research on the morphology of fracture position, with a primary focus on image binarization processing. The process of extracting the outlines of bone joints necessitates a high degree of precision. However, the application of binarization often results in significant distortion, thereby hindering its ability to meet the established requirements. Furthermore, given the numerous levels present within the grade division and the minimal disparity between them, some of which are indistinguishable to the unassisted eye, it is imperative to extract the key components of the bone joint to ensure optimal recognition accuracy. In this paper, an automatic grading diagnosis system for osteoarthritis KL is proposed. This system is designed by improving the edge detection algorithm and adopting a reasonable classification algorithm concurrently. This study addressed the issue of subjectivity in manual evaluation, and its high accuracy and efficiency suggest that it can serve as a reliable computer-aided decision-making tool for clinical OA diagnosis, with the potential to replace manual grading.

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